

Tom Cantellow

tcantellow@gmail.com | [Linkedin](#) | [Website](#) | [Portfolio](#)

EXPERIENCE

Senior Executive Officer

Mar. 2026 – Present

Strategic Social Research Team, Department for Environment, Food & Rural Affairs

Newcastle/London, UK

- Leading development of a reproducible analytical pipelines toolkit for social research, applying statistical methods to with reusable R templates and guidance, forming a strategic framework for embedding reproducible analysis.
- Continuing provision of expert advice on AI governance, survey analytics, and responsible deployment of Large Language Model (LLM) solutions across social research teams.

R&D Fellow

Aug. 2025 – Mar. 2026

Strategic Social Research Team, Department for Environment, Food & Rural Affairs

Newcastle, UK

- Designed and delivered Retrieval Augmented Generation (RAG) LLM chatbot prototypes grounded in survey data; foundational architecture subsequently reused by other departments, accelerating delivery of parallel programmes.
- Engineered end-to-end data pipelines using Databricks and Azure, embedding RAP principles across Python, R, and SQL workflows. Led cross-government knowledge exchanges on RAG and LLM approaches for use in analysis.

Research Associate

Apr. 2023 – Dec. 2024

University of Bristol

Bristol, UK

- Developed Net Zero Vulnerability Index using machine learning and eigenvector-based indices to rank UK Local Authorities by low-carbon transition risk; extended to parliamentary constituencies via IPPR funding.
- Presented findings across 8+ government departments; research informed a ministerial briefing to the DESNZ Secretary of State. Published peer-reviewed journal article, policy briefing, public engagement article, technical methods note, interactive webmap dashboard, and submitted parliamentary evidence.

Teacher

Oct. 2020 – May 2025

University of Bristol

Bristol, UK

- Designed and delivered quantitative analysis lectures teaching Excel, R, and data visualisation to diverse undergraduate cohorts, adapting communication for technical and non-technical audiences.
- Mentored students on analytical approaches, dataset construction, and statistical methods, building analytical capability and promoting best practices in data handling and presentation.

EDUCATION

University of Bristol

Bristol, UK

PhD in Human Geography (Quantitative Methods)

Oct. 2020 – Nov. 2024

- Authored a 58,000-word peer-reviewed PhD thesis analysing UK voting behaviour and how the operationalisation of neoliberalism shapes inequalities, political values, and personal financial wellbeing.
- Integrated large datasets (British Election Study, Land Registry, Census) using mixed-effects regression and Extract-Transform-Load workflows with data quality assurance and validation.

University of Sheffield

Sheffield, UK

MSc, Distinction, in Applied GIS – Ian Masser Prize for best dissertation

Sep. 2017 – Sep. 2018

University of Lincoln

Lincoln, UK

BA (Hons), 2:1, in International Relations

Sep. 2012 – May 2016

AWARDS & RECOGNITION

Anna Kukkola Memorial Award nomination for exceptional professional development (DEFRA, 2026).

Bristol STAR Award nomination for contributions to inclusive research culture (University of Bristol, 2023).

Ian Masser Prize for best MSc dissertation, University of Sheffield (2018).

Sheffield Postgraduate Scholarship awarded £10,000 for MSc tuition fees (2017).

TECHNICAL SKILLS

Programming: R, Python, SQL, HTML/CSS, LaTeX

Platforms & Tools: Databricks, Azure, Power BI, RStudio, Jupyter, QGIS, ArcGIS, Git, Excel.

Analytical Methods: Machine learning, mixed-effects modelling, regression, decision trees and random forests, spatial econometrics, ETL and RAP pipelines, RAG LLM development, survey methodology, data governance, research design.

Datasets: ACLED, BES, BSA, BRES, CHES, EPCs, IMDs, Scottish Census, SAE, UK Census, UKHLS,

UK Land Registry & many other large administrative datasets.